

A JAVA TRAVEL DIARY · IN TWO ACTS

being the working papers of one

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SENIOR SOFTWARE ENGINEER · JAVA · DISTRIBUTED SYSTEMS · AWS

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Shahjahanpur, UP · open to relocate ♦ open to Senior SWE postings & correspondence

-47%

P90 LATENCY, PRICING FAN-OUT
1.10 s → 0.59 s

+33%

TRAVELER CONVERSION
POST RE-ARCHITECTURE

-43bps

COST-PER-TRANSACTION
PAYMENTS LEDGER

20 → 5

ON-CALL PAGES PER SHIFT
AFTER GUARDRAILS

CHAPTER · I

I.

The Year His Pagers Went Quiet

In which the narrator, appointed custodian of four Java microservices along the booking funnel, teaches a pricing engine to breathe, a payments ledger to forget its double-counts, and a weekly on-call roster to sleep through the night.



S D E II — *at* Amazon Travel · Hotels, Flights & Insurance

JUL 2023 · SEP 2025 · BENGALURU

Owner of four Java microservices across the booking funnel; principal engineer on the pricing, payments, deployment, and on-call fronts of a global traveler platform.

I

THE PRICING FAN-OUT. Re-architected the traveler's pricing call upon `CompletableFuture.allOf` with bulkhead `ExecutorService` pools — cut p90 from **1.10 s to 0.59 s** (-47%) and lifted traveler conversion by +33%.

II

THE IDEMPOTENT LEDGER. Wrote a payment webhook upon SQS and Step Functions, keyed by DynamoDB dedupe entries — reclaimed **-43 bps CPT** and silenced the double-capture incidents for good.

III

THE CDK REWRITE. Rewrote six legacy pipelines in Java CDK — **halved rollbacks**, and pulled MTTR down by **a full day**.

IV

THE SILENCED PAGER. Dragged the weekly on-call from **20+ pages to ~5**, config-hygiene from 5-6 → 0 through automated guardrails; resolved **150+ incidents**, and stopped ~30 bps of funnel-churn on insurance attach.

Act I · The Apprentice Years

being the earlier chapters, the catalogue of instruments, and two works cited

CHAPTER · V

S D E I — *at* Amazon Travel

JUL 2022 · JUL 2023

THE FIRST COMMISSION. Authored REST controllers and @Repository DynamoDB DAOs upon Spring DI; steered an auto-insurance renewal flow to **+60% renewal completion**. Held the unit-test line above **80% coverage** with JUnit 5 and JaCoCo gates in CI.

CHAPTER · VI

S D E Intern — *at* Amazon Travel

MAY 2021 · FEB 2022

THE APPRENTICE PAPERS. Saw a GraphQL façade retired into pure Java; engineered a *One-Click Renewal* upon Guice DI; kept the test ledger above **90% coverage** throughout.

◆ THE CATALOGUE OF INSTRUMENTS ◆

JVM

Java 17, Spring Boot, Spring MVC, Spring DI, Guice, Maven, Gradle

CONCURRENCY

ExecutorService, CompletableFuture, ConcurrentHashMap, ScheduledExecutor, bulkheads

AWS

Lambda, DynamoDB, SQS, Step Functions, API Gateway, AppSync, CloudWatch, ECS, CDK (Java)

QUALITY

JUnit 5, Mockito, JaCoCo, integration harnesses, contract tests

FRONT-END

React (SSR & CSR), Redux, TypeScript, Vitest

DATA

DynamoDB single-table, idempotency keys, dedupe queues

OPS

runbooks, alarm hygiene, SEV pages, on-call rotations

METHOD

trunk-based, code-review discipline, postmortems, SLO-first

◆ WORKS CITED ◆

Redis Server · *in Java*

[GITHUB.COM/PRAXSTACK/REDIS-SERVER-JAVA](https://github.com/praxstack/redis-server-java)

A from-scratch study of the RESP2 protocol, event-loop I/O, and a hand-rolled hashmap engine — built in the evenings, benchmarked in the small hours.

SET 154K RPS GET 195K RPS INCR 128K RPS 42 TESTS
~78% OF C IMPL

Markdown Viewer · *App*

[GITHUB.COM/PRAXSTACK/MARKDOWN-VIEWER-APP](https://github.com/praxstack/markdown-viewer-app)

A zero-config reader with a live plugin architecture and a keyboard-first command palette, carried by a lean gzipped payload and no runtime baggage.

155+ VITEST TESTS <100KB GZIPPED ZERO RUNTIME DEPS

Open-source: 3 PRs merged into [danielmiessler/Fabric](#) (42k-star Go AI-tooling framework) — Bedrock bearer-token auth, dynamic region fetching, and a streaming-deadlock fix.

EDUCATION · COLOPHON

M.C.A., Computer Science — *MNIT Prayagraj*

2019 — 2022 · grade-book 9.25 / 10

B.Sc. (Hons.), Computer Science — *Bhaskaracharya College, DU*

2015 — 2018

RECOGNITION · POSTSCRIPT

AMAZON PAY MERCHANTS — CATEGORIES CHAMPION, 2023. Commended for stewardship of the merchant-category domain.

TRUST-BUSTER SPRINTS. Resolved **100+ trust-buster bugs** across travel quality sprints — small cuts, quietly healed.